Jaime Gómez

PhD Student in mathematics at EPFL
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Phone number: +41 78 243 38 67, **Citizenship:** Spanish



Education

EPFL *Lausanne, Switzerland*
PhD studies in mathematics *2023–*

Supervisor: Prof. Joachim Krieger.
Research group: Partial Differential Equations.

ETH Zurich *Zurich, Switzerland*
Master's degree in mathematics *2021–2023*

Master's thesis supervisors: Prof. Alessio Figalli & Dr. João Pedro Gonçalves Ramos.
Thesis title: Stability of Faber-Krahn inequalities in time-frequency analysis.
Graduated with an ETH medal for an outstanding master thesis and a GPA of 5.84/6.

Universidad Autónoma de Madrid *Madrid, Spain*
Bachelor's degree in mathematics with programming minor *2017–2021*

Bachelor's thesis supervisor: Prof. Ana Vargas Rey.
Thesis title: The Fourier transform and Hausdorff measures. (ES)
Graduated with honors on the bachelor's thesis and a GPA of 8.7/10.

Research

Stability of the Faber-Krahn inequality for the STFT *Inventiones Mathematicae*
With André Guerra, João P.G. Ramos & Paolo Tilli. *2024*

Uniform stability of concentration inequalities and applications *Proc. London Math. Society*
With D. Kalaj, P. Melentijević & João P.G. Ramos. *2025*

Multisoliton solutions and blow up for the L^2-critical Hartree equation *ARMA*
With Tobias Schmid & Yutong Wu. *2025*

Effective threshold dynamics for the cubic nonlinear Klein-Gordon equation in three dimensions *ArXiv:2610.41718*
With Joachim Krieger. *2026*

Sharp stability of the Faber-Krahn inequality for Bergman spaces and log-subharmonic functions *In preparation*
With D. Kalaj, P. Melentijević & João P.G. Ramos. *2026*

Refereed for: American Journal of Mathematics and zbMATH Open.

Awards and honors

2024 **Awarded** the ETH Medal for an outstanding master thesis.

2017 **Awarded** the extraordinary high school award by the Castille and Leon autonomous region.

2017 **Awarded** for being among the best 100 grades in the selectivity exam in the university district of Salamanca.

2016 **5-th place** in the physics olympiad in Salamanca.

2016 **6-th place** in the mathematics olympiad in Salamanca.

Work and teaching experience

Teaching Assistant *EPFL*
TA in the courses “Analyse avancée I”, “Analyse avancée II”, “Introduction to dynamical systems”, “Calculus of variations” and “Harmonic analysis” *2023-2027*

Teaching Assistant *ETH Zurich*
Author of the official script for the course “Functional Analysis II” *2022, 2024*

Student tutoring *Madrid, Spain*
Private lessons for high school students *2020–2021*

Languages and skills

Language skills

- **Spanish:** Native.
- **English:** Fluent.
- **Italian:** Intermediate.
- **French:** Basic.

Musical and piano studies *Salamanca, Spain*
Professional conservatory of Salamanca *2009–2015*

Academic projects

QED magazine *Madrid, Spain*
Writer and proofreader *2021–*

QED (ES) is an association of mathematics students in Universidad Autónoma de Madrid. They regularly publish issues of a magazine where students write articles about mathematics.

Stability of Isoperimetric inequalities *ETH Zurich*
Supervisor: Prof. Alessio Figalli. *2022*

Semester paper on stability of isoperimetric inequalities.

The Faber-Krahn inequality for the short time Fourier transform *ETH Zurich*
Supervisors: Prof. Alessio Figalli & Dr. João Pedro Gonçalves Ramos. *2023*

Semester paper on signal processing and the theoretical limitations of energy concentration on the time-frequency plane.

Lecture notes
Lecture notes for several courses in the master’s degree in mathematics.

Programming skills and projects

Programming skills

Programming minor (C, 1 yr), with mathematics-oriented courses (R, Matlab, 1/2 yr, SageMath, 1 yr).

- **Intermediate/advanced:** C, GDScript, SageMath, LaTeX .
- **Basic:** R, HTML & CSS, C++, C#, Swift, Matlab, JavaScript.

LaTeX2InD

Zurich, Switzerland

Creator

2023

LaTeX2InD is the first ever script designed to include LaTeX functionality in Adobe InDesign. Open source, JavaScript.

Interests

I enjoy various sports, mostly endurance-based, such as running and cycling (both road and gravel), and I love using my bike to explore. I particularly enjoy the problem-solving aspects of climbing, and I love doing it outside, specifically lead climbing. In winter, skiing is my sport of choice. I also enjoy cooking, computer graphics (in particular water simulations à la Tessendorf), aviation, traveling, and I love cinema and photography. I always take my camera with me whenever I travel anywhere.

Personal information

Citizenship: Spanish (European Union), with a **Swiss residence permit** (B).
Born in October 1999 in Salamanca, Spain.
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